Gray-Zone Maritime Control in the First Island Chain

A Networked Escalation Game

Workshop 1 Report (Deliverable 2)

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1 Setting and research questions

China and the United States compete over maritime presence and access in the First Island Chain, including the Senkakus, Taiwan Strait, Luzon Strait, Scarborough and Second Thomas Shoals, and Paracels, below the threshold of war. China presses its maritime claims through persistent coast-guard and militia presence backed by PLAN ships; U.S. presence and freedom-of-navigation operations contest excessive maritime claims and support access (Helmus et al., 2024; Green et al., 2017). The model represents Blue with destroyers (DDG) and Red with navy (PLAN) and coast-guard/militia (CCG) hulls; LCS are not a separate modeled type. Every two days, each side chooses where to station ships and whether to patrol (P) or assert (A). Assertion abstracts actions such as shadowing, close maneuvering, querying and warning, and U.S. freedom-of-navigation operations; it is not an identical real-world tactic for both sides.

Strategic interaction is central: control depends on the local force ratio, incidents depend on both sides' presence and posture, war costs both players, and each side has incomplete observations of the opposing fleet.

RQ1. Control and escalation. How do incident-risk budgets affect control and war probability, and which budget pairs are Nash equilibria in the restricted budget-choice game?

RQ2. Policy performance. How do policies derived from game-theoretic models compare with random and rule-based policies in payoff, control, and war probability against a common set of opponents?

RQ3. Robustness and redesign. Do the best Blue policies and designs under nominal weather also perform best in expected and worst-case payoff across weather scenarios, against the tested Red policies?

Outcomes are estimated over 200 seeds per configuration: each player's payoff, the fraction of games ending in war, and Blue's average value-weighted control index $\bar{c}_{\mathsf{B}} = \tau^{-1} \sum_{t=1}^{\tau} (\sum_v V_v^{\mathsf{B}} c_v^{t+1}) / (\sum_v V_v^{\mathsf{B}}) \in [-1, 1]$. RQ3 also compares scenario-weighted and worst-scenario payoff. Repeated-game restraint is a supporting analytical example, rather than a separate simulation claim.

Response to proposal feedback.

Feedback	Revision
Make the model explicit	Define state, actions, observations, transitions and payoffs (§3).
Formulate Q1 as a MIP	Use a myopic allocation MIP; risk budgets replace informal player types (§3.4).
Deepen the research questions	Add a budget-choice game, policy comparisons, and resilience analysis (§4).
Specify uncertainty and redesign	Four weather scenarios and three Blue designs address the robustness feedback (§3).
Supply parameters and evidence	Mark parameter values as assumptions; report analytical results and 77 configurations with 200 seeds each (§4).

2 Literature and current practice

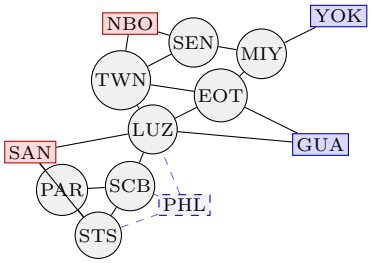
Helmus et al. (2024) document Chinese gray-zone tactics, including persistent presence, resupply obstruction and water cannon; Green et al. (2017) discuss deterrence and escalation management. CUES provides an encounter-management reference (WPNS, 2014). These motivate the action categories and asymmetric hull types, rather than calibrating numerical parameters.

The model combines structured gray-zone gaming (Wasser et al., 2019), game-based patrol allocation (Shieh et al., 2012), network interdiction (Washburn and Wood, 1995), and escalation through shared risk (Schelling, 1966). Its contribution is a computable general-sum game for comparing policies and designs. Control, incident hazards and weather probabilities are modeling assumptions, not empirical estimates.

3 Game formulation

3.1 Agents, network and state

Players are $\mathcal{I} = \{\mathsf{B}, \mathsf{R}\}$: Blue has 10 DDG; Red has 8 PLAN and 6 CCG hulls. The undirected graph $G = (\mathcal{V}, \mathcal{E})$ contains home ports $H_{\mathsf{B}} = \{\text{YOK}, \text{GUA}\}$, $H_{\mathsf{R}} = \{\text{NBO}, \text{SAN}\}$ and eight contested zones $\mathcal{C}$ (Fig. 1). One period is two days; moving one edge takes one period.



Sequence within each period

1. Nature updates regional weather; players receive observations.
2. Endurance rules identify ships that must return or refit.
3. Both sides choose moves and postures simultaneously.
4. Moves resolve within fleet movement budgets; endurance updates.
5. Control and incidents update; escalation rises or falls.
6. Payoffs accrue; play stops at war or after 60 periods.

Figure 1: Network and sequence of play. Squares are ports; circles are contested zones. Dashed links and PHL are the additional access assumed in design D1.

The state is $s^t = (x^t, d^t, c^t, \ell^t, w^t)$:

Symbol	Meaning	Symbol	Meaning
$x^t_{k,v}$	Type-k ship count at node v	d^t_s	Ship s's remaining endurance
c^t_v	Control: -1 Red to $+1$ Blue	ℓ^t	Escalation: 0 to $L = 5$ (war)
w^t_r	Calm, rough or storm in region r	V^i_v	Player i's value for zone v

3.2 Actions, constraints, information and uncertainty

Each free ship stays or moves to an adjacent legal node; each occupied zone receives posture $p_{i,v} \in \{\text{P}, \text{A}\}$. For directed traversals (u, v) of the undirected edges, let $f^t_{k,uv} \in \mathbb{Z}_{\geq 0}$ count moving ships. For $k \in \mathcal{K}_i$,

$$x^{t+1}_{k,u} = x^t_{k,u} + \sum_{v \in \delta(u)} f^t_{k,vu} - \sum_{v \in \delta(u)} f^t_{k,uv} \qquad \text{(ship conservation)},$$

$$\sum_{v \in \delta(u)} f^t_{k,uv} \leq x^t_{k,u}, \qquad \sum_{k \in \mathcal{K}_i, (u,v)} f^t_{k,uv} \leq F_i \qquad \text{(available ships; movement cap)}.$$

Here $\delta(u)$ denotes adjacent nodes and $F_{\mathsf{B}} = 8$, $F_{\mathsf{R}} = 10$ ship movements per period. This is a logistical movement proxy, not a fuel inventory. Ships with $d_s \leq \text{dist}(v, H_i)$ receive return priority; one full period in port resets endurance to D_k. Opponent ports and storm-closed nodes cannot be entered; ships in storm zones hold. The engine executes return moves first up to F_i; unexecuted moves hold, and free ships use the remaining budget.

Information. Each side sees its own fleet and public control, escalation and weather. At each observed node it draws a Bernoulli detection with probability $q_{i,v}$: 0.95 with a ship present, 0.6 with an adjacent ship or own-port coverage, and 0.2 otherwise. Red has a 0.6 floor at four coastal zones before the bad-weather multiplier 0.7. Conditional on the state, detections are drawn independently. Success reveals exact counts by type and posture; a miss retains the last observation. Initial counts are known. This is a last-seen estimate, not a Bayesian belief distribution.

Weather. Scenarios $\omega = (\text{calm}, \text{nominal}, \text{typhoon}, \text{extreme})$ have weights $\pi = (0.25, 0.40, 0.25, 0.10)$. Regional three-period storms start with probabilities $p_s = (0, 0.02, 0.06, 0.12)$ when no storm is active; otherwise rough-sea probabilities are $p_r = (0, 0.2, 0.35, 0.5)$. Storms freeze local control and prevent incidents; rough seas halve CCG effectiveness.

3.3 Transitions and payoffs

Local strength and control respond to presence and posture:

$$S_{i,v} = \sum_{k \in \mathcal{K}_i} w_k \alpha^{\mathbf{1}[p_{i,v}=\text{A}]} x_{k,v}, \quad \phi_v = \frac{S_{\mathsf{B},v} - S_{\mathsf{R},v}}{S_{\mathsf{B},v} + S_{\mathsf{R},v} + \kappa}, \quad c^{t+1}_v = (1-\lambda)c^t_v + \lambda\phi_v. \tag{1}$$

Strength uses post-move counts. Here $w_{\text{DDG}} = w_{\text{PLAN}} = 1$, $w_{\text{CCG}} = 0.6$ (0.3 in rough seas), $\alpha = 1.5$, $\lambda = 0.3$, and $\kappa = 1$.

Incidents and escalation. At a non-storm zone, an incident has probability $\rho_v = 1 - e^{-h_v}$,

$$h_v = \big(h_0 + h_{\mathrm{A}}^{\mathsf{B}}\mathbf{1}[p_{\mathsf{B},v} = \mathrm{A}] + h_{\mathrm{A}}^{\mathsf{R}}\mathbf{1}[p_{\mathsf{R},v} = \mathrm{A}]\big)\min\{S_{\mathsf{B},v}, S_{\mathsf{R},v}\}(1 + \ell^t/2). \tag{2}$$

Red's assertion coefficient is weighted by its PLAN/CCG strength mix. Each incident raises escalation by one, capped at L; with no incidents, escalation falls by one with probability μ (floor zero).

Payoffs. Let m_i^t count executed moves, $\sigma_{\mathsf{B}} = 1$, $\sigma_{\mathsf{R}} = -1$, and C_i be the per-period design cost ($C_{\mathsf{R}} = 0$). If ℓ^{t+1} is the post-turn escalation,

$$g_i^t = \sigma_i \sum_{v \in \mathcal{C}} V_v^i c_v^{t+1} - \gamma m_i^t - \eta \ell^{t+1} - C_i, \qquad u_i = \sum_{t=1}^{\tau} \delta^{t-1} g_i^t - W\mathbf{1}[\mathrm{war}]. \tag{3}$$

The horizon τ is the earlier of $T = 60$ and the first turn ending at $\ell = L$. Different zone values and a shared war cost make this a general-sum game.

Zone	TWN	SEN	MIY	EOT	LUZ	SCB	STS	PAR
V_v^{B}	10	6	4	4	6	5	5	2
V_v^{R}	10	6	3	4	5	7	6	6
c_v^1	0	0.2	0.3	0.2	0.1	−0.2	0	−0.4

Table 1: Assumed zone values and initial control.

Parameter group	Assumed values
Payoff and escalation	$\delta = 0.98$, $\gamma = 0.1$, $\eta = 1$, $W = 200$, $\ell^1 = 1$, $\mu = 0.25$
Incident hazard	$h_0 = 0.006$; assertion: Blue 0.03, Red PLAN 0.036, CCG 0.012
Endurance and incident cost	$D_{\mathrm{DDG}} = 10$, $D_{\mathrm{PLAN}} = 8$, $D_{\mathrm{CCG}} = 6$ periods; $\Lambda = 20$

3.4 Decision models: myopic allocation and template game

The *MIP-BR* policy optimizes a myopic surrogate against last-seen counts and postures, rather than the full dynamic game. Let $y_{s,w}$ select a legal destination $w \in A_s$ for free ship s, and let $b_{v,(n,p)}$ select a count vector n and posture p at zone v. Tabulating $\phi_v(n,p)$ and $\rho_v(n,p)$ makes the allocation problem linear:

$$\begin{aligned}
\max_{y,b,z} \quad & \textstyle\sum_{v\in\mathcal{C},(n,p)} a_v(n,p) b_{v,(n,p)} - \gamma \sum_{s,w\neq \mathrm{loc}(s)} y_{s,w} - 1000z \\
\text{s.t.} \quad & \textstyle\sum_{w\in A_s} y_{s,w} = 1 && \forall s \quad \text{(one destination)}, \\
& \textstyle\sum_{s,w\neq\mathrm{loc}(s)} y_{s,w} \le F_i^{\mathrm{rem}} && \text{(remaining movement budget)}, \\
& \textstyle\sum_{(n,p)} b_{v,(n,p)} = 1 && \forall v \quad \text{(one zone option)}, \\
& \textstyle\sum_{(n,p)} n_k b_{v,(n,p)} = \bar{x}_{k,v} + \sum_{s:\,k(s)=k} y_{s,v} && \forall v,k \quad \text{(consistent counts)}, \\
& \textstyle\sum_{v,(n,p)} \rho_v(n,p) b_{v,(n,p)} \le \varepsilon_i(1 - \ell^t/L) + z && \text{(incident-risk allowance)}, \\
& y, b \in \{0,1\}, \quad z \ge 0.
\end{aligned}$$

Here $a_v = V_v^i \lambda H \sigma_i \phi_v - \Lambda \rho_v$, $H = [1 - \delta(1-\lambda)]^{-1}$ weights control persistence, $\bar{x}$ counts prescribed return/refit positions, and $F_i^{\mathrm{rem}} = \max(0, F_i - \text{prescribed return moves})$. Variables for illegal destinations are absent. The risk sum estimates *expected incidents this turn*, not war probability. Penalized slack z makes the budget soft. Budgets $\varepsilon_i \in \{0.05, 0.15, 0.40\}$ represent conservative, moderate and aggressive play.

Template game. Each player chooses one of 14 templates: seven regional fleet splits times two postures. With calm weather and $\ell = 1$, payoffs are $M_i = \sigma_i \sum_v V_v^i \phi_v - \Lambda \sum_v \rho_v$. Dominance, Nash equilibria, maximin and correlated-equilibrium LPs apply to this restricted game Γ. Its equilibrium and maximin policies are then tested in the simulator, alongside random, rule-based, greedy and MIP-BR policies. Equilibrium guarantees do not transfer automatically to the simulator.

3.5 Blue redesigns

D1: dispersed basing adds an assumed PHL port and links to SCB/LUZ/STS (cost 0.3/period). **D2: sensing** adds a 0.6 detection floor in contested zones, including bad weather (cost 0.5/period). **D3: hotline** raises μ to 0.5 and halves assertion hazards when $\ell \ge 3$ (zero assumed cost). Two unitary actors, last-seen estimates and myopic decisions remain limitations.

4 Preliminary results

Analytical results use the restricted template game. Simulations use the same 200 seed identifiers per configuration. Unless specified, Red uses MIP-BR with $\varepsilon_R = 0.15$ in nominal weather.

4.1 Supporting analysis: incentives and repeated-game restraint

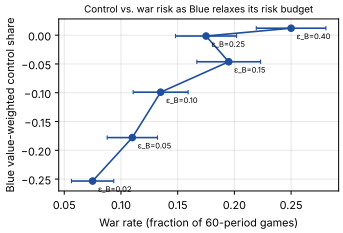
Fix Blue's center-heavy and Red's south-heavy splits. Normalizing (P, P) to $(0, 0)$ gives $(P, A) = (-9.61, 1.20)$, $(A, P) = (0.28, -10.45)$ and $(A, A) = (-11.41, -11.33)$ (Blue, Red). This chicken-type game has two pure equilibria. In the mixed equilibrium, assertion probabilities are 0.576 (Blue), 0.135 (Red), with payoffs $v_B = -1.30$, $v_R = -6.02$. A welfare-maximizing correlated equilibrium avoids mutual assertion.

In an *infinitely repeated, perfectly monitored version of this fixed subgame*, patrol followed by permanent mixed-equilibrium punishment after any assertion is sustainable when $\delta \geq G_i/(G_i - v_i)$, with deviation gains $G_B = 0.28$, $G_R = 1.20$. Thresholds 0.18 and 0.17 support restraint at $\delta = 0.98$ in this auxiliary game, not necessarily in the finite, partially observed simulator.

4.2 Template-game equilibria

Strict dominance reduces 14×14 templates to 2×4. Pure equilibria are (Blue center-heavy assert, Red south-heavy patrol), payoffs $(0.39, -9.27)$, and (Blue center-heavy patrol, Red center/south-heavy assert), $(-14.44, 2.71)$. The tested policy selects the first by total payoff. A separate zero-sum control game (Red receives minus Blue control payoff) has both sides assert in the Center, value -2.26.

4.3 RQ1: control, war risk and budget choices



B\R	0.05	0.15	0.40
0.05	$-194, 199$	$-303, 315$	$-308, 318$
	0.10	0.11	0.12
0.15	$18, -36$	$-144, 101$	$-192, 140$
	0.15	0.20	0.21
0.40	$74, -80$	$-83, 16$	$\mathbf{-151, 39}$
	0.14	0.25	**0.34**

Figure 2: Left: control–war tradeoff with Red's budget fixed at 0.15. Right: budget-game payoffs (u_B, u_R) with war rates beneath; payoff s.e. approximately 13. Bold marks the pure equilibrium of the estimated payoff matrix.

Raising Blue's budget from 0.02 to 0.15 lifts control from -0.25 to -0.05 and war rate from 0.07 to 0.20; 0.40 adds only 0.06 control. The estimated matrix has a unique pure equilibrium at $(0.40, 0.40)$ and its highest war rate (0.34). Some comparisons (318 versus 315) are within noise. A lower total payoff alone does not establish a prisoner's dilemma.

4.4 RQ2: performance against common opponents

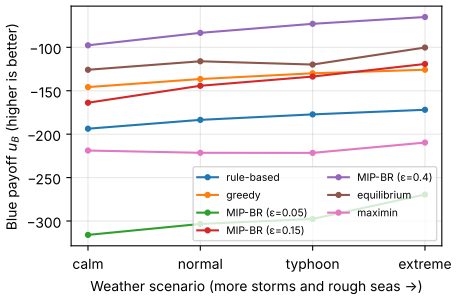
Blue policy	Red rule-based	Red MIP-BR	Red equilibrium	Worst tested payoff
Random	-283	-319	-164	-319
Rule-based	-122	-184	35	-184
Greedy	-70	-137	65	-137
MIP-BR	-75	-144	53	-144
Equilibrium	-72	$\mathbf{-116}$	21	$\mathbf{-116}$
Maximin	-170	-221	-35	-221

Table 2: Blue payoff under nominal weather, baseline D0; reported s.e. 1–7. Control and war-rate results for every matchup remain in the repository's policy results.

Against Red MIP-BR, the equilibrium-derived policy improves control from -0.05 to -0.03 and halves war rate (0.20 to 0.10) relative to Blue MIP-BR. Its best worst payoff covers only the tested opponents. Maximin has lower payoff but war rates at most 0.08. Across 1,524 decisions, greedy matched the MIP optimum in 67%. Over those 30 games, greedy exceeded the risk allowance in 208 decisions (14%) and MIP in 37 (2.4%), consistent with its penalized slack.

4.5 RQ3: weather robustness and resilient designs

For policy or design x, let $J(x, \omega)$ be its simulated mean payoff under scenario ω. We compare $\sum_\omega \pi_\omega J(x, \omega)$ and $\min_\omega J(x, \omega)$. The latter is the worst scenario mean, not the worst individual run or worst possible opponent.



Design	Calm	Nom.	Typh.	Extr.	Exp.	Worst
D0	−164	−144	−134	−119	−144	−164
D1	−153	−141	−131	**−98**	−137	−153
D2	−181	−164	−149	−136	−162	−181
D3	**−140**	**−118**	**−111**	−102	**−120**	**−140**

Figure 3: Left: Blue policy payoffs by weather. Right: Blue design payoffs with MIP-BR on both sides (reported s.e. approximately 7); D0 baseline, D1 basing, D2 sensing, D3 hotline.

Weather. Most tested policies do worst in calm seas: rough seas weaken Red's CCG hulls and storms pause incidents. Baseline MIP-BR's war rate falls from 0.24 (calm) to 0.10 (extreme). MIP-BR with $\varepsilon_B = 0.40$ and the equilibrium-derived policy rank first and second in all four scenarios. The highest-payoff policy also has war rates of 0.12–0.29, making sensitivity to $W = 200$ important.

Designs. D3 has the best expected and worst-scenario payoff; in nominal weather it cuts war rate from 0.20 to 0.05 and increases Red's payoff from 101 to 144. Its zero cost is an assumption. D1 barely changes Blue's nominal payoff, reduces Red's from 101 to 16, and raises war rate to 0.28. D2 shows no measurable benefit even in the reported zero-cost check. Red recomputes actions under its fixed MIP-BR policy; these experiments do not re-solve equilibrium policy selection after redesign.

Limits of inference. Numerical values are assumptions. The findings are comparisons within the model, not calibrated forecasts of conflict. War rates depend on hazards, de-escalation and war penalties. Small payoff gaps should be treated cautiously given Monte Carlo error; formal comparisons require uncertainty estimates for payoff differences. The engine's return-move cap can also make executed positions differ from the MIP's prescribed fixed positions when returns alone exceed the cap.

5 Prototype

`wargame_gui.py` supports interactive play, batch runs and turn-by-turn traces. Users select policies, design and weather, then step through periods or run many games. `Game.step` implements the sequence in Fig. 1; `MIPBRPolicy` solves the allocation model; `MetaGame` computes the template-game analyses.
`experiments.py all` regenerates the reported experiments; see `prototype/README.md`. Learning agents, Bayesian state estimates and multi-period policies remain to be developed.

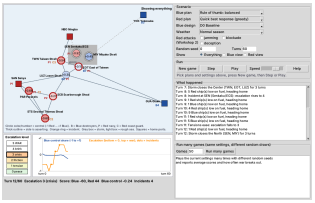


Figure 4: GUI after 12 turns; storm in the North.

6 Plan for Workshop 2

- **Model and validation:** add Bayesian beliefs and multi-period decisions; test return-budget edge cases, report risk slack, and sweep V, h, W, μ by $\pm 50\%$.
- **Learning and adaptation:** evaluate reinforcement learning for Red and fictitious-play or no-regret updates in the template game; measure convergence, restraint and exploitation of rule-based play.
- **Attacks and defenses:** evaluate the coded blockade, jamming and false-contact attacks; add an attacker responding to Blue's posture. Compare randomized patrols, D1 and D2 using control and war rate.
- **Resilient designs:** test D0–D3 and combinations under attacks and long typhoons, price the hotline, and allow Red to adapt its policy choice to each design.

Milestones: Nov 3, beliefs and attacks; Nov 17, learning; Dec 1, redesigns; Dec 8, final report.

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